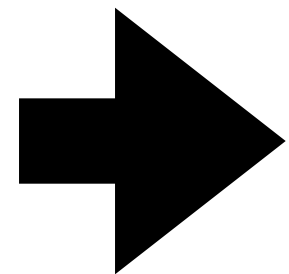


Introduction to Data Science with Python

Dr. Scott Ladenheim

Glassdoor Best Jobs in America, 2022

	Job Title	Median Base Salary	Job Satisfaction	Job Openings
#1	Enterprise Architect	\$144,997	4.1/5	14,021
#2	Full Stack Engineer	\$101,794	4.3/5	11,252
#3	Data Scientist	\$120,000	4.1/5	10,071



- US Bureau of labor statistics: jobs expected to grow 36% from 2023 to 2033
- In the Top 5 Fastest Growing Six Figure Jobs in 2025 (*Forbes*)

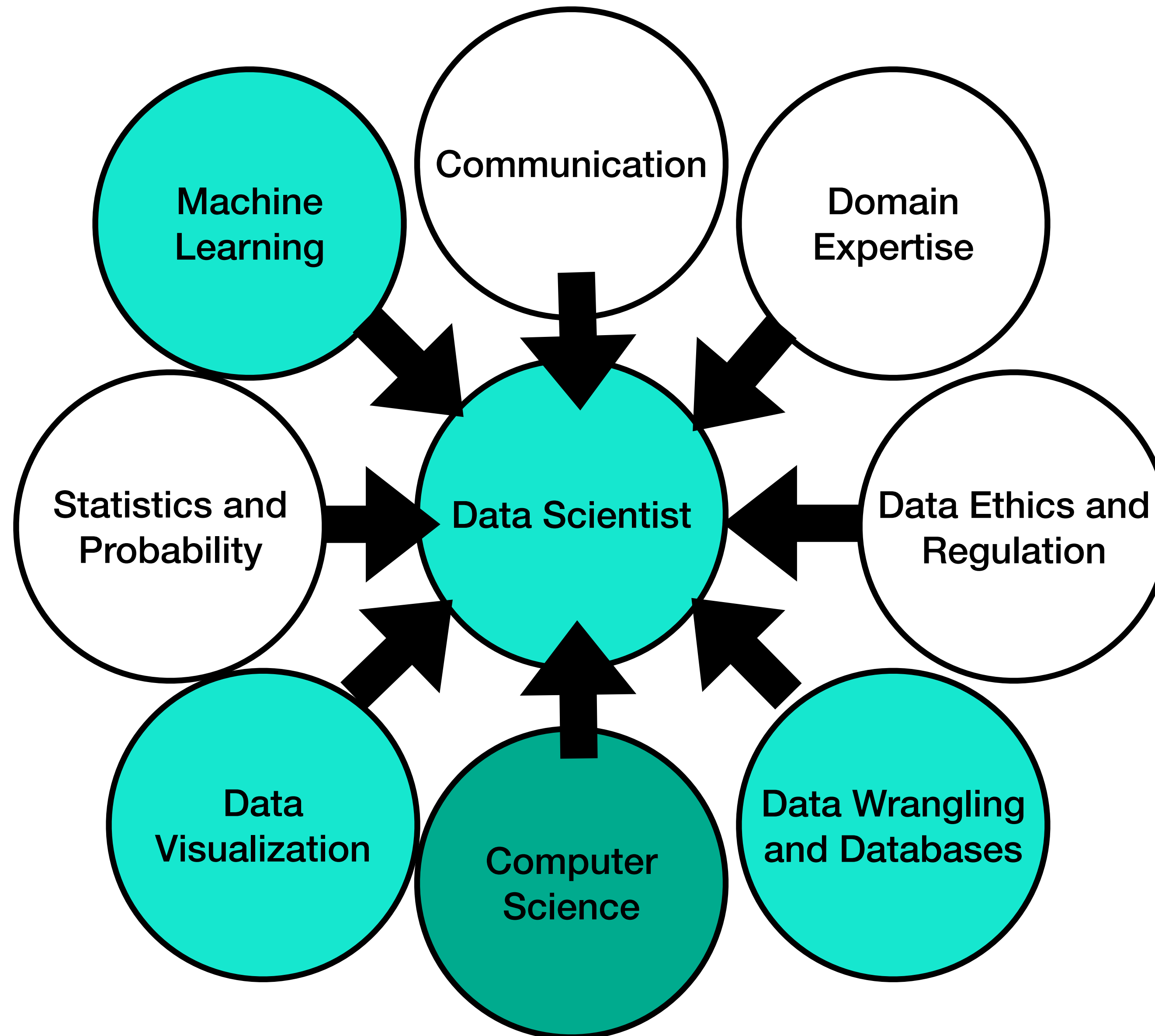
What does a data scientist do that is so satisfying?

- "What a real data scientist does is take data — it can be large, small, from a variety of sources — and interpret it for your client or your employer."
-Michael Walker, President, Data Science Association
- **Building** the pipeline of code that takes data and analyzes it
- **Revealing patterns** through the use of statistics and machine learning
- **Communicating findings effectively** through visualization and reports

Satisfied Data Scientists

- "My favorite part of the job is creating a plan of attack for a new problem: envisioning the steps necessary to take raw data and put it into a form that gives meaningful, actionable knowledge to the client. That's when I usually discover the most interesting new technologies, techniques, and models..."
-Andrew Graczyk, Senior Data Scientist, NNData
- "I enjoy seeing the final product. The result of our work and the result of our algorithms are often things that are really, really cool."
- Lotem Peled, Head Data Scientist, Gong.io
- "We're creating a better future." - Michael Walker, President, Data Science Association

Our slices of data science in this course



The Importance of Python in Data Science

- You aren't going to analyze all that data by hand - you need **code**
- Programmers benefit from the code of others, and Python has the best "**ecosystem**" of other programmers writing data-related code you can use
- Some Python **libraries** of code commonly used by data scientists:
 - numpy, scipy - for fast, accurate mathematical computation
 - pandas - for organizing data in tables
 - scikit-learn - for easily deploying basic machine learning (ML)
 - Pytorch/Tensorflow/Keras - for making your own deep neural networks - advanced ML
 - seaborn/plotly - for good-looking plots

Sketch of Semester

Python
Weeks 1-7

Midterm 1 week 7

Intro CS topics
Week 8

Machine Learning
Weeks 9-10

Complexity & Algs
Week 11

SQL & Midterm 2
week 12

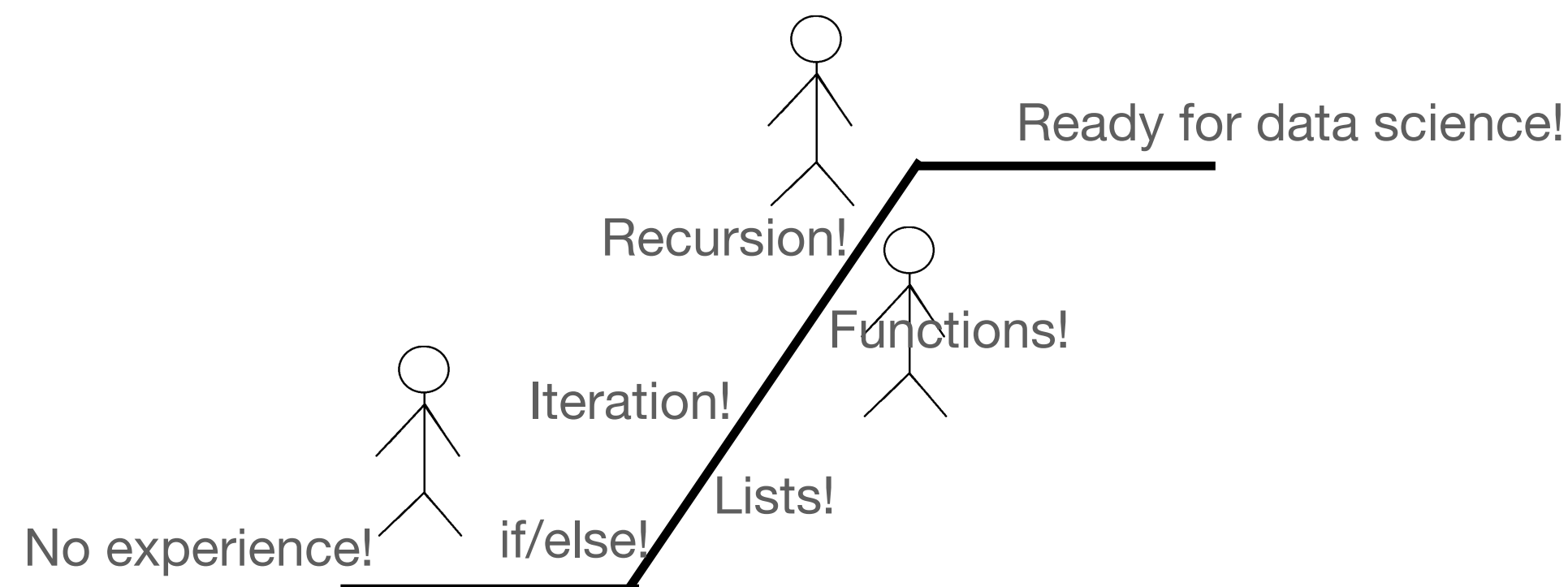
More topics weeks 13-14

Lightning talks & review
week 15

Final Exam

Skills Covered in DS110: Python Programming

- Experts, bear with us as everybody gets caught up to speed on this stuff!
- Programming essentials: storing & manipulating data, conditions, loops, etc
 - No prior experience needed, but newbies may need to work harder at first!
- Slightly more advanced general topics: object-oriented programming, data structures, recursion
- Different people will "get on the escalator" and start learning new things at different times
- Note that DS100 is a less steep learning curve



Skills Covered in DS110: Handling Data

- Working with large datasets (not to be confused with "Big Data" though):
 - pandas library for organizing data
 - Reading and writing files
 - Regular expressions to look for patterns

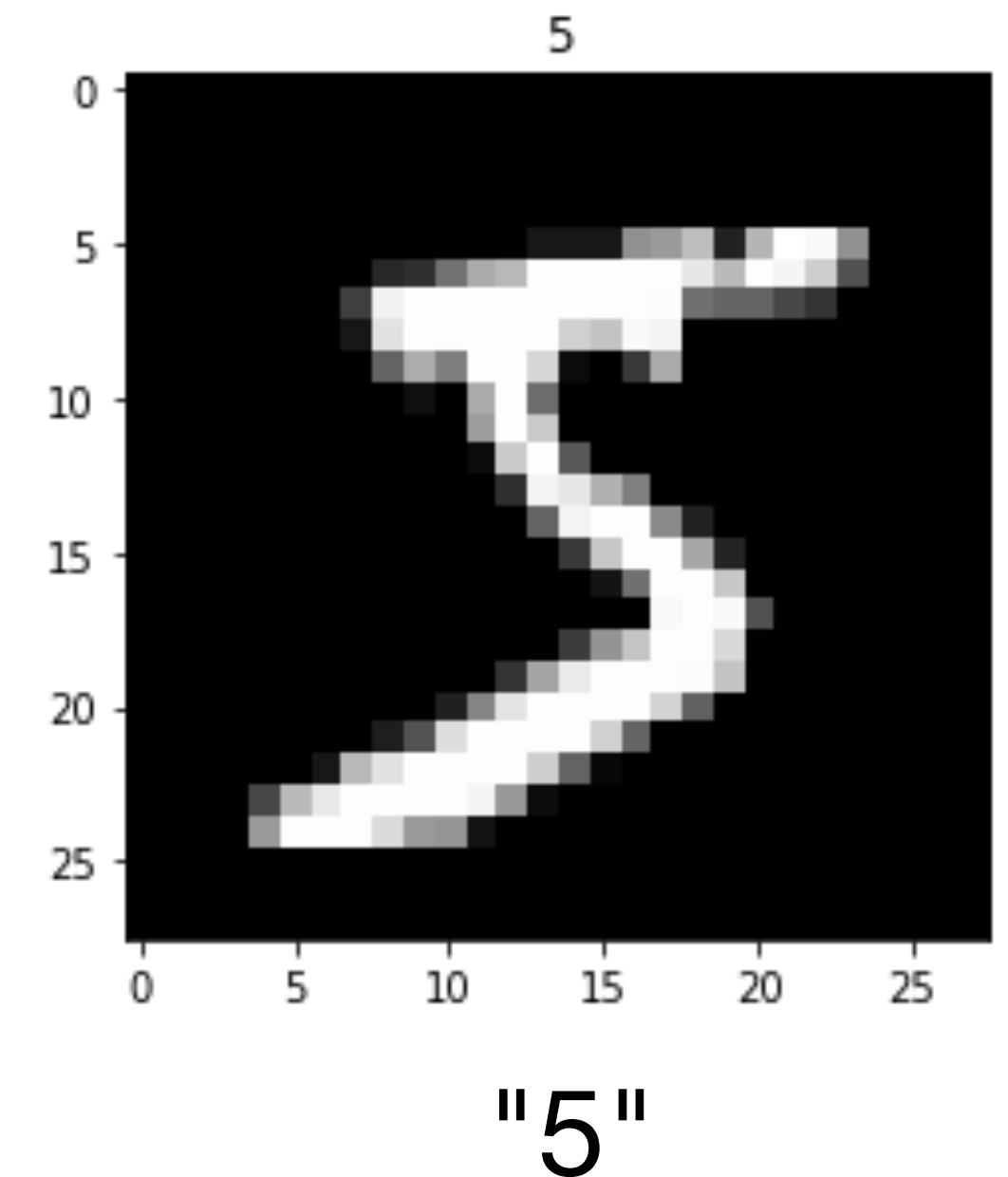
```
In [33]: data
```

```
Out[33]:
```

	Area Abbreviation	Area Code	Area	Item Code	Item	Element Code	Element
0	AF	2	Afghanistan	2511	Wheat and products	5142	Food
1	AF	2	Afghanistan	2805	Rice (Milled Equivalent)	5142	Food
2	AF	2	Afghanistan	2513	Barley and products	5521	Feed
3	AF	2	Afghanistan	2513	Barley and products	5142	Food
4	AF	2	Afghanistan	2514	Maize and products	5521	Feed
5	AF	2	Afghanistan	2514	Maize and products	5142	Food
					Millet and		

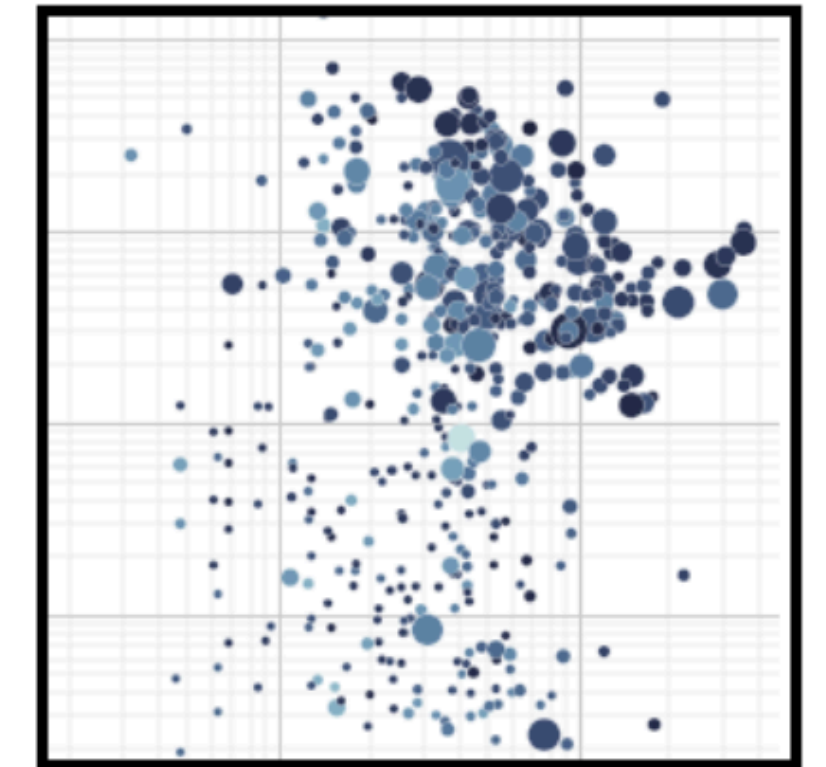
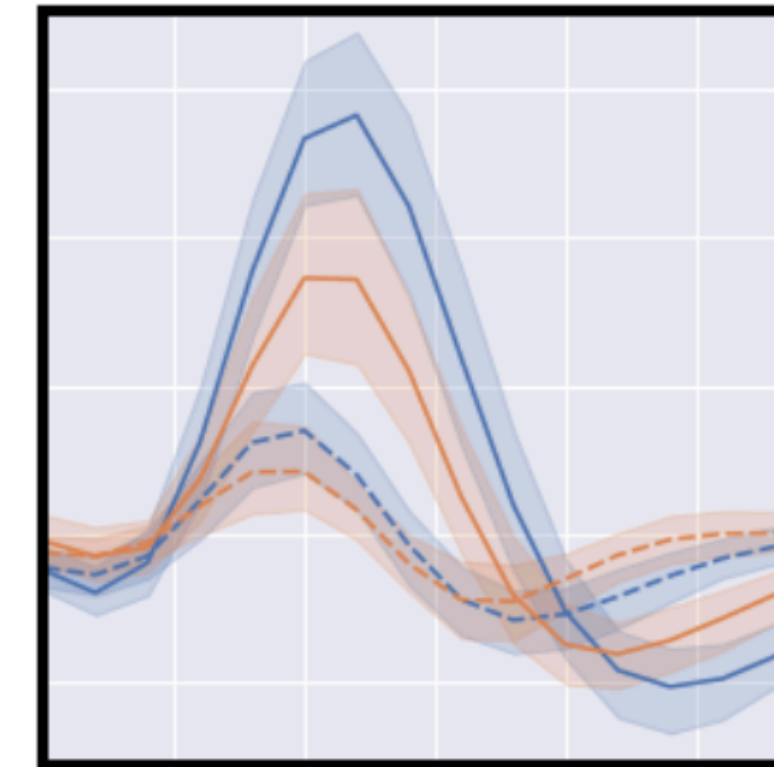
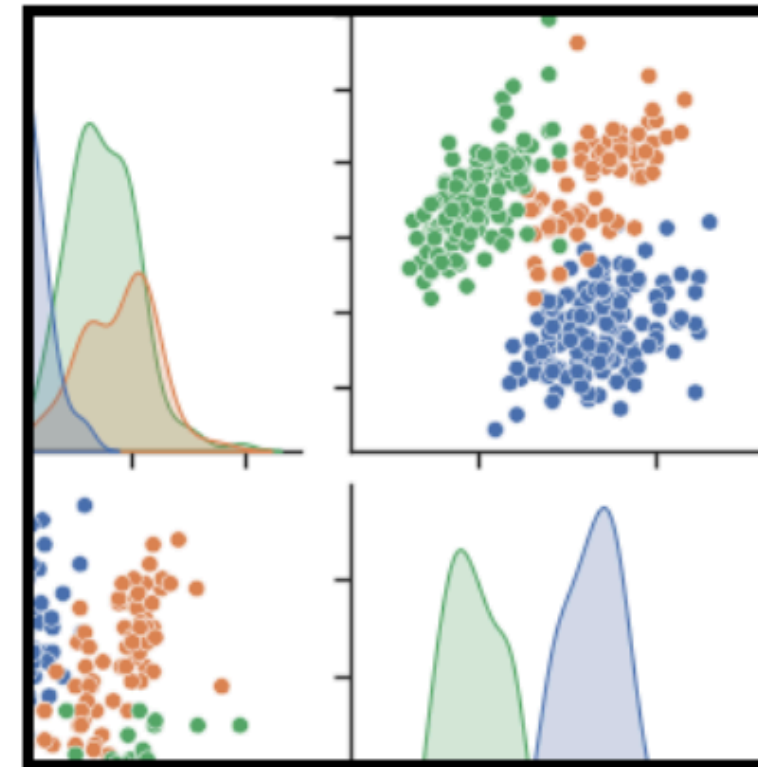
Skills Covered in DS110: Machine Learning

- Classifiers that can learn to classify new inputs on the basis of examples
 - k-nearest neighbors
 - Decision trees
 - Decision forests
- Regression: machine learning that predicts values
- Embeddings: applying machine learning to text



Skills Covered in DS110: Visualization

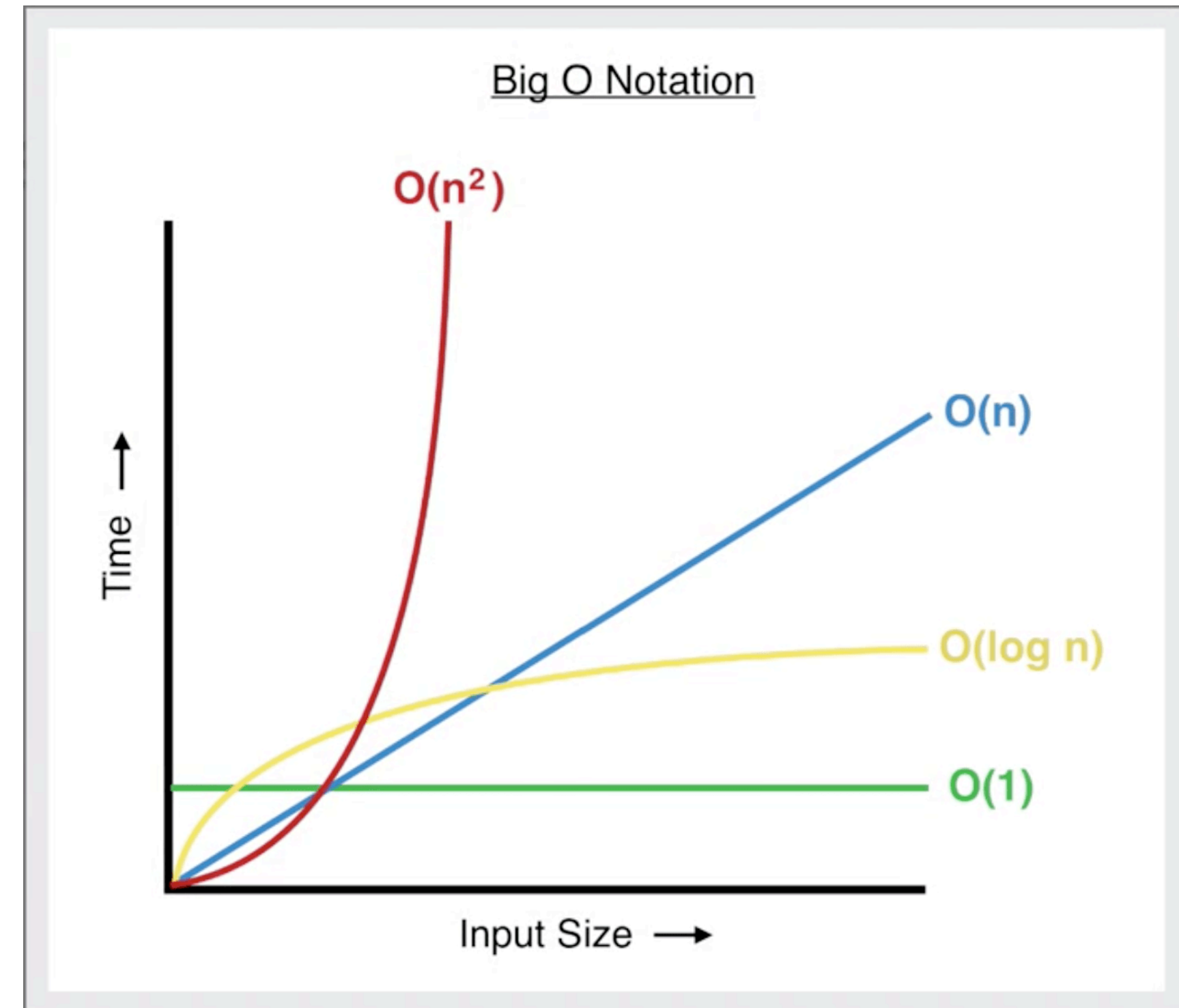
- Libraries for plots and graphs
 - matplotlib
 - seaborn
- Anybody know which is generally considered better, a pie chart or a bar chart?



<https://seaborn.pydata.org/>

Skills Covered in DS110: Faster Code

- When you work with a lot of data, your code needs to be fast
- We'll study:
 - How to analyze the running time of code ("big-O notation")
 - How to make use of multiple processors to (theoretically, maybe) go faster
 - How to use a profiler to find bottlenecks in running time
 - A few other tricks for going faster



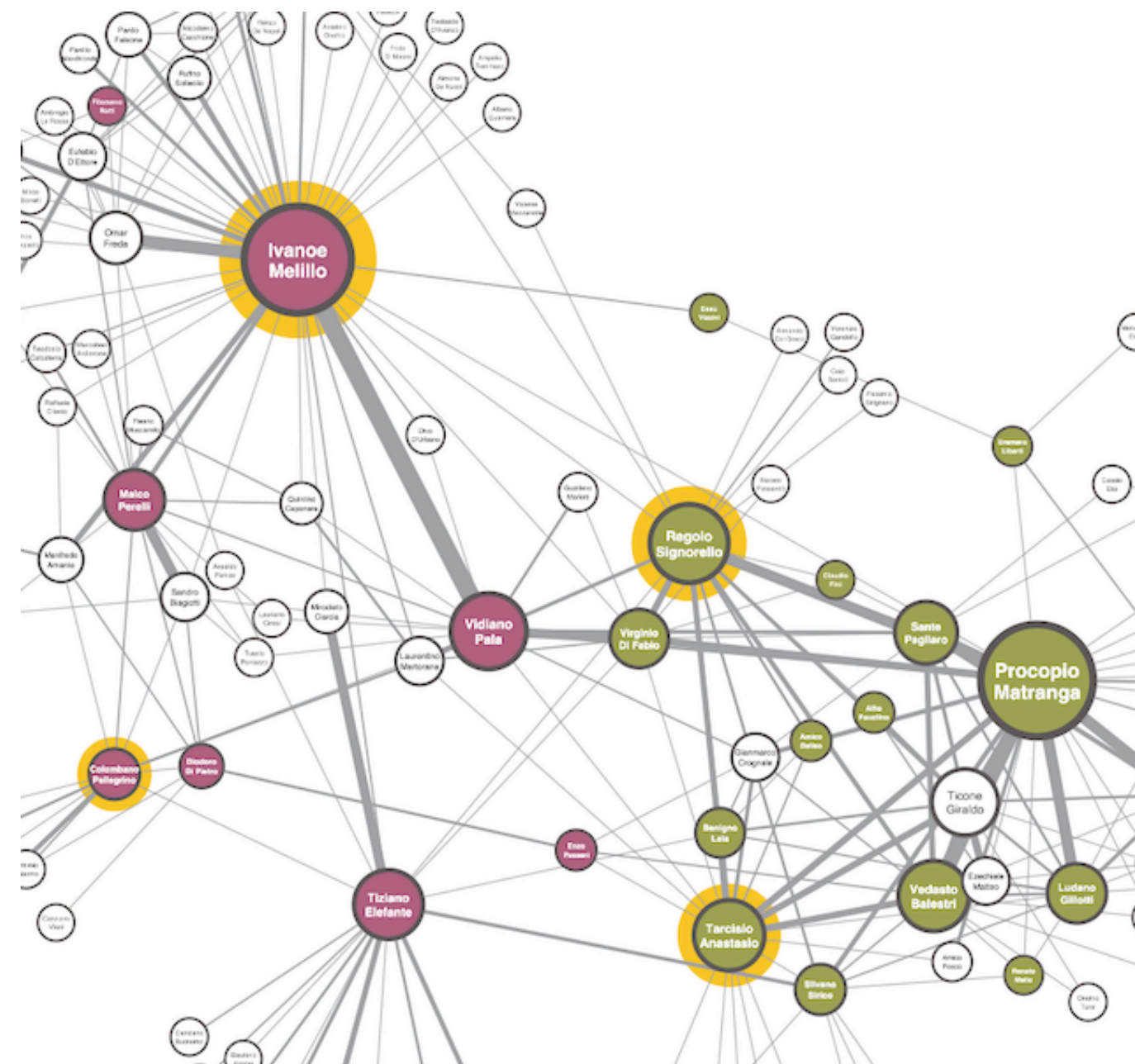
```
def vectorized_mult(n):  
    return np.array(range(n)) * 2  
  
%time vectorized_mult(12345678)
```

CPU times: user 1.13 s, sys: 147 ms, total: 1.28 s
Wall time: 1.28 s

Assorted other one-lecture topics

- SQL - intro to a query language for databases
- Graphs and centrality - essentials of analyzing social networks

```
SELECT title, edition, copyright FROM titles  
WHERE copyright > '2016'
```



Primary Learning Goals

1. **Learn how to program in general:** variables, iteration, data structures, functions, debugging
2. **Learn core Python syntax, keywords, and programming patterns** (how do I use variables *in Python*? Do iteration *in Python*?...)
3. **Learn the basics of applied machine learning**
4. **Gain exposure to a broad array of tools and ideas in data science** (but not the math, that's DS120-121-122)
5. **Open-ended learning through a deep dive into the dataset of your choosing**

Grade breakdown

- In-class quizzes (10%)
- Individual assignments (15%, lowest dropped)
- Midterm exam 1 (15%)
- Midterm exam 2 (20%)
- Recitation participation (5%)
- Final exam (35%)

Quizzes and Kahoots



- Quizzes are **unannounced ahead of time** in an effort to encourage attendance.
 - 1 question about the current lecture
 - 1 per week
 - Asked in class, answer in Blackboard
- Kahoots are not-for-credit quizzes for review - download the Kahoot! app to play easily

Homework and Late Policy

- Out and due on Wednesdays (usually!)
- Receive and turn in Jupyter notebooks
- No copied work, but discussion is permissible with acknowledgement
- **AI use must be documented**
- 5 late days to use over the course of the semester, no more than 2 per assignment
 - Work won't be accepted past the second consecutive day or if 5 days are up



The screenshot shows a Jupyter Notebook titled "Lecture20RandomForests.ipynb". The interface includes a menu bar with "File", "Edit", "View", "Insert", "Runtime", "Tools", and "Help", along with a "Last saved at 2:06 PM" timestamp. Below the menu bar, there are tabs for "+ Code" and "+ Text". The notebook content consists of a text cell followed by a code cell. The text cell contains the following text: "Decision forests aren't as inspectable as decision trees, but you can. Here is a scikit-learn use of decision forests on images, the LFW per great fit, because the decisions don't lend themselves to individual p". The code cell contains the following Python code:

```
from sklearn.datasets import fetch_lfw_people
from sklearn.ensemble import RandomForestClassifier

faces = fetch_lfw_people(min_faces_per_person = 100)
```

On Generative AI for the Homework

- A variety of AIs exist that can write code and answer questions about it for you, including the free GPT 4o-mini, Github copilot, and even one built in to Colab
- *Good use of AI:* Asking how to do something in general. "How do I convert a string to an integer?"
- *Good use of AI:* "Can you explain this code I saw in lecture? [code]"
- *So-so use of AI:* Asking where your bug is. It helps you learn if you figure this out yourself, but it's true that some bugs are not that interesting. Make sure to ask *why* a change is needed if you don't understand.
- *Bad use of AI:* Accepting autocompleted code that you don't understand.
- *Bad use of AI:* Getting a whole solution to a problem.

Students who overuse AI on the homework are not going to do well on the exams, worth 70% of the grade!

Midterms and Final

- We balance a *permissive AI policy* with *in-class exams* to see what you really know - good tools are nice, but so is working efficiently & succeeding in interviews
- 2 midterms - week 7, week 12 - and one final
- Roughly one multiple choice question per lecture, plus 2-3 paper-and-pencil programming problems

Our generative AI experiment: Practice Problems

Midterm 1 Topics ▾

☒ If	☐ Lists	☒ While	☐ Boolean Operators	☐ For Loops
☐ Nested Loops	☐ Functions	☐ Sets or Dictionaries	☐ matplotlib	☐ numpy

Midterm 2 Topics ▾

☐ pandas	☐ Regular Expressions	☐ Files and Exceptions
☐ Object-oriented Python	☐ Recursion	☐ Graphs

Generate Problem

- <https://huggingface.co/spaces/klgold/ds110probs>
- Need your BU username and a 4-digit PIN from Blackboard ("AI PIN")
- Generate random programming problem on topics of your choice - can enter your solution for feedback

Mini-Project

- There will be one opportunity for extra credit, in HW8
- Analyze a dataset of your choosing
- The best will be invited to give a lightning talk 12/8

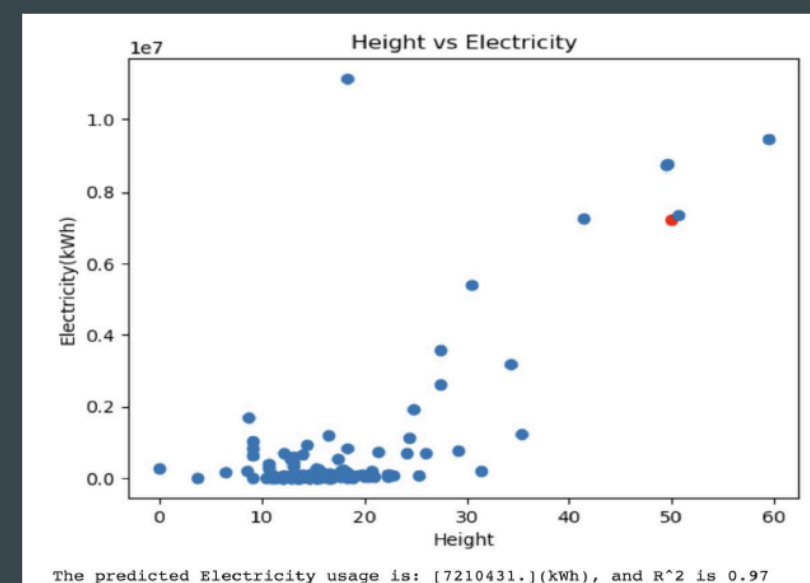
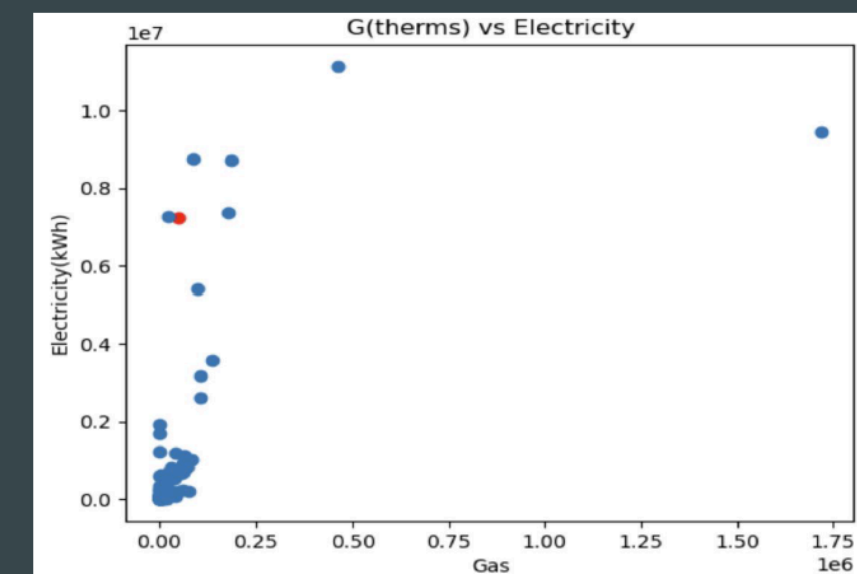
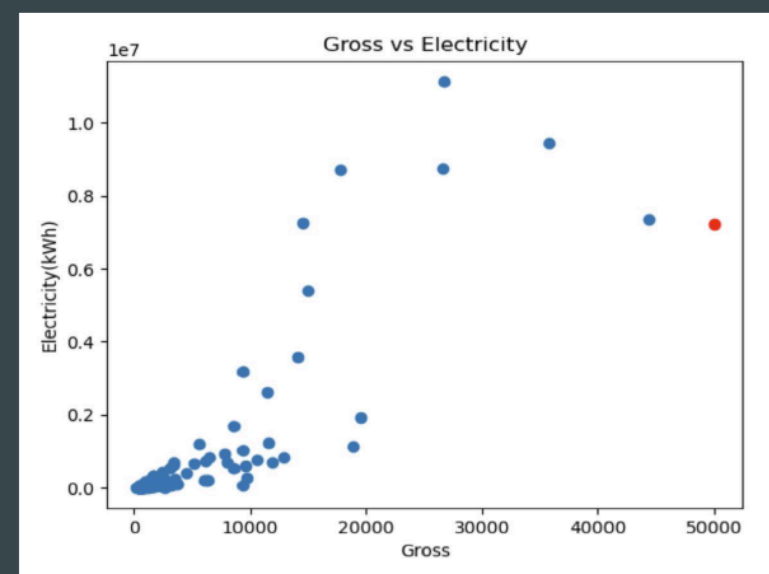
Conclusion

Final program: predictor(gas, gross, height)

To the right is the result when we run our program with:

- Gas = 50000
- Gross = 50000
- Height = 50

The predicted Electricity usage is 7,210,431 kWh and R^2 is 0.97.



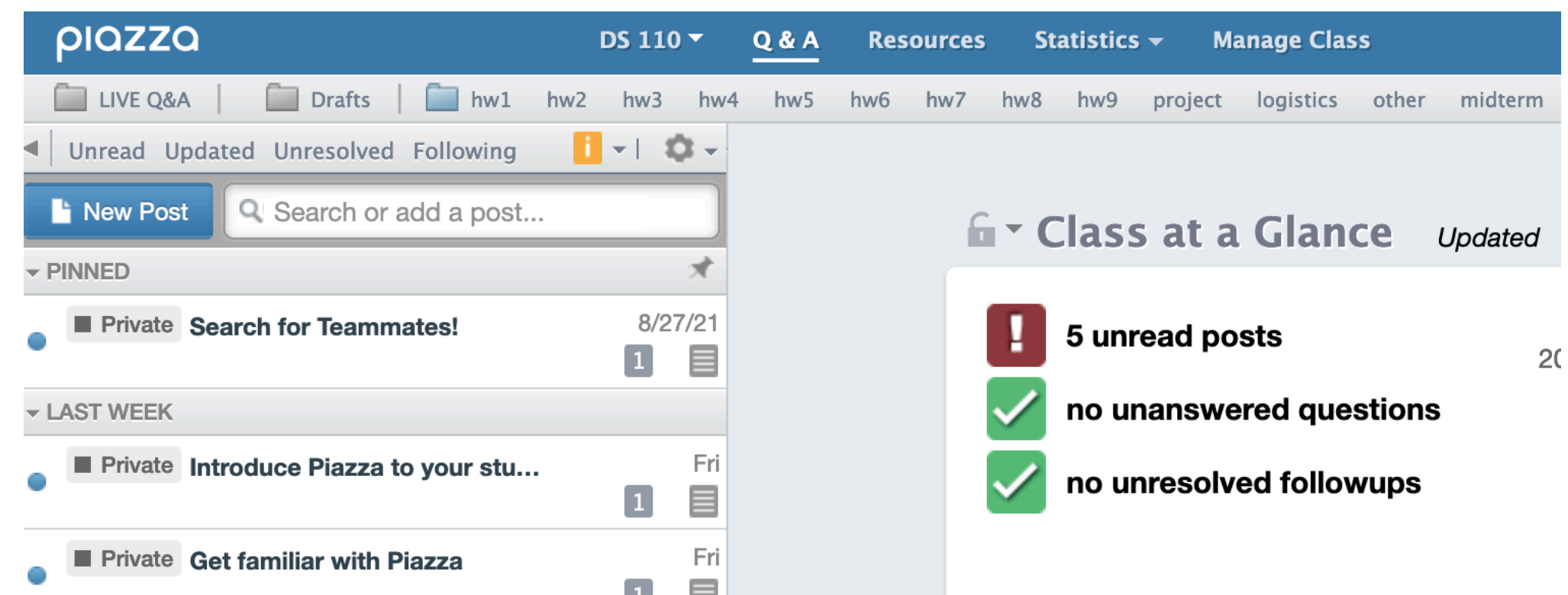
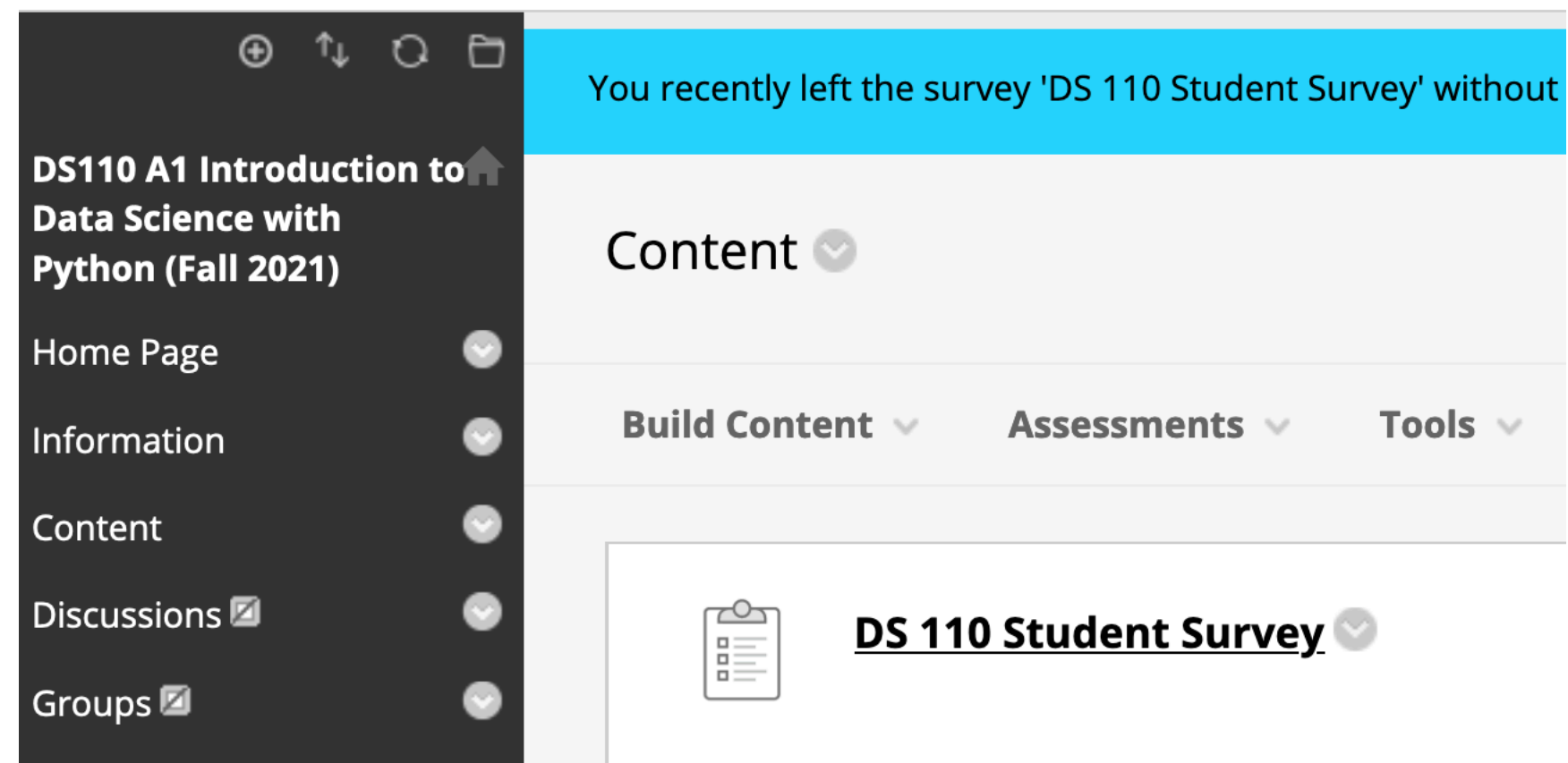
The predicted Electricity usage is: (7210431.1)(kWh), and R^2 is 0.97

Predicting Boston University Electricity Usage

Vijay Fisch, Tony Zhang, Rheona Mehta, Trinity Yacoub

Blackboard and Piazza

- Assignments will go out and be graded on Blackboard
 - learn.bu.edu
- Announcements and assignment Q&A over Piazza - **sign up!**
 - **<https://piazza.com/bu/fall2025/ds110/>**



Tutoring resources

- A free tutor is available through the Educational Resources Center
- Sign up online with TutorTrac: erctutor.bu.edu

Google Forms survey (link on Blackboard->Content)

Python experience?

- ☐ None
- ☐ A little on the side to support some project or coursework
- ☐ At least one intro CS course or equivalent
- ☐ A lot

Office hours on Piazza

- <https://piazza.com/bu/fall2025/ds110/staff>

DS 110: Introduction to Data Science with Python

Boston University - Fall 2025

 Syllabus   

Course Information

Staff

Resources

Staff Office Hours

	Kevin Gold	 M 3-4:30pm, F 10-11:30am  CDS 1406
	Scott Ladenheim	 MW 2-3PM  CDS 1545

[TA OH will also go here]

Use Piazza to Q&A with me and the staff

- <https://piazza.com/bu/fall2025/ds110/>

 question @141   

Project Grades

When can we get our project final grades approximately?

project

 Edit good question | 0

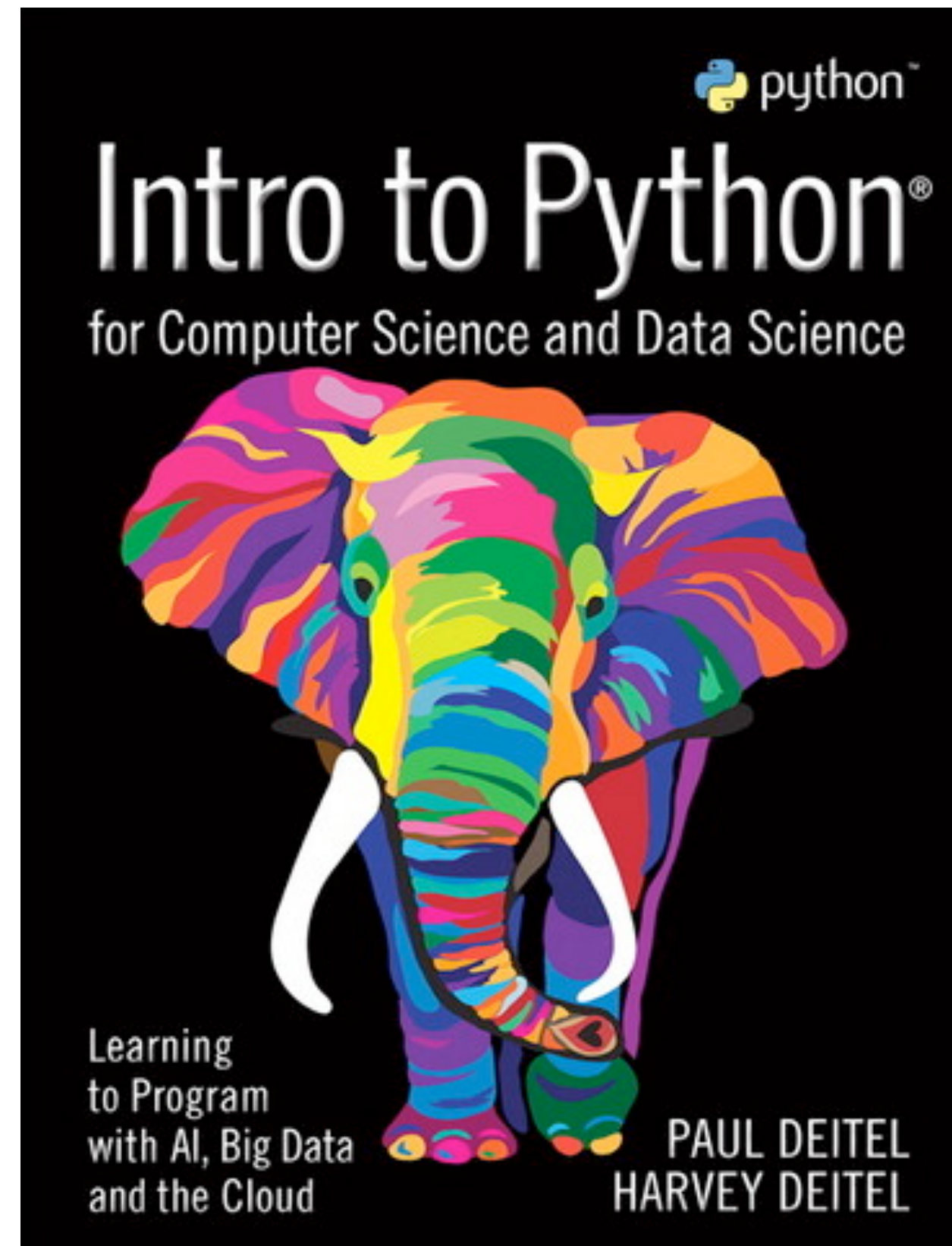
 **the instructors' answer**, *where instructors collectively construct a single answer*

Approximately Thursday.

 Edit good answer | 6

The recommended textbook

Useful if you want more practice problems or a different explanation



Jupyter notebooks

- Google colab: <https://colab.research.google.com/>

▼ Sample problem: digits

Our dataset that we will try to successfully fit will be a relatively low resolution, and there's nothing in the image besides the digit. This is a good dataset to learn from. The data consists of 8x8=64 brightness values for each pixel. For a new digit, a trained learner should be able to produce a correct classification.

```
[ ] from sklearn.datasets import load_digits
    import matplotlib.pyplot as plt
    digits = load_digits()
    print(digits.data.shape) # Examples x 64 pixels
```

```
(1797, 64)
```

Welcome, everyone!